

P3: TL retention after EN continuation (nanochat, TL-39 then WikiText-103) (#307342)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has one author.

Pre-registered on:
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1) Have any data been collected for this study already?

It's complicated. We have already collected some data but explain in Question 8 why readers may consider this a valid pre-registration nevertheless.

2) What's the main question being asked or hypothesis being tested in this study?

This study was designed after P2 Gate U/V unblinding on 19 August 2026 (#306935). It is a post-P2 reverse-direction preregistration, not confirmation of P2, and does not amend #306780 or #306935.

After a newly trained P3-specific Tagalog nanochat parent (TL0; fresh weights; frozen P1.1 WikiText-TL-39 train documents; new P3 Tagalog tokenizer) passes P0-T, does English continuation on WikiText-103 raise Tagalog held-out BPB more than matched extra Tagalog?

$C_{tl} = TL \text{ val_bpb_full}(B2) - TL \text{ val_bpb_full}(B1)$. Predict $C_{tl} \geq 0.01$ at d20.

$G_{en} = EN \text{ val_bpb_full}(B2) - EN \text{ val_bpb_full}(B1)$. Predict $G_{en} \leq -0.01$.

B3 is a 50/50-document mix, not mitigation. Report $C_{tl}(B3)=TL(B3)-TL(B1)$ and $G_{en}(B3)=EN(B3)-EN(B1)$ plus realized document/byte/token shares.

P0-T, before any English train token, at BOTH d8 and d20: TL0 Tagalog val_bpb_full is ≥ 0.01 below untrained same-depth and below a train-fitted add-1 UTF-8 byte unigram on the same val. If either depth fails either floor, do not train B1/B2/B3; report blocked parent-eligibility.

Do not load P1.1 or P2 weights. Same P3 Tagalog 32768 BPE on every arm. Not CORE/chat/SFT.

3) Describe the key dependent variable(s) specifying how they will be measured.

Primary DVs: Tagalog val_bpb_full on frozen P1.1 reconstructed_article_70_15_15 val SHA256 4d51644b84d05050bfc8c515079e60f6e437082b6cce2122e9ed00e7b1db2b1c; English val_bpb_full on official WT103-raw val document manifest SHA256 874dec29844b3d46fc39e5479ee2dc4b3ba37309d9baf3bba4b5654697f3ae3b (not the raw HF archive hash). Same P3 Tagalog 32768 BPE. BPB = mean token NLL / $(\ln(2) * \text{mean UTF-8 bytes per token})$, official evaluate_bpb, T=2048, BOS-bestfit, full val. meta.val_bpb is not a DV.

Six child cells: B1/B2/B3 x {TL, EN}. C_{tl} and G_{en} as in Q2. During Gate U, evaluate B0 English val_bpb_full once with the same locked evaluator; record it descriptively; exclude it from C_{tl} , G_{en} , and B3 contrasts. P0-T covers B0 Tagalog.

After val seal, B2 d20 only, two LEGACY EXTERNAL HOLDOUTS (not virgin P3 tests): EN WT103-raw test document manifest SHA256 2bccabc020cbb8d09273ccdc42ed926957b83824ca767c96fb588041b8d434e (used in P2 Gate V); TL P1.1 test.jsonl SHA256 3bd193458f4c494d84dae345548c0c01cb6cd7275e98d6ed39a41d517a093baf (used in P1.1 and P2). Unused for P3 tok/train/val. Not test-set C_{tl}/G_{en} . Do not reuse 1.164768 or P2 test numbers.

4) How many and which conditions will participants be assigned to?

No human participants. Pin 92d63d4e8bb4df75c3b71618f31ddde2378b2bcd; tok_train -> base_train -> evaluate_bpb; T=2048; CORE off.

TL0: fresh weights; P1.1 Tagalog train SHA256 2b0474c5700dc1eba14def572aa23cc227e4c59c10c2de3ce6b7bda75d137687; NEW P3 Tagalog 32768 BPE; d8 and d20; $N_{TL0} = \text{ceil}(3 * T_{tl_train}/65536)$. B0 = freeze TL0 d20. Never load P1.1/P2 .pt.

B1 extra Tagalog. B2 official WT103-raw train. B3 50/50 documents frozen before any TL0 val (seed 42, sha256-sort, K=min, interleave). $D_{\text{phase2}} = 294 * 65536 = 19267584$. Fresh pinned Muon+AdamW (load_optimizer=False; no inherited optimizer/scheduler/scaler/resume). LR peak = $0.3 * TL0$ peak. Warmup < N (default 14 if $14 < N$).

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

P0-T at d8 and d20 before any English train token. Operators see PASS/BLOCKED only. If either depth fails either floor, do not train B1/B2/B3.

Primary: C_{tl} and G_{en} on the six child cells. Gate U B0 English val is descriptive only. B3 formulas as in Q2; not mitigation.

One seed: point estimates; no CI or significance test. $|\text{delta}| < 0.01$ is not a ranking. Do not retune mix/D/cutoff/test from P2. 294-step budget, 0.01 cutoff, and 50/50 B3 are mechanical carry-forward, not tuned to P2 magnitudes. Replay, FilBench, EWC, CORE, SFT are exploratory.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

Drop a document only if null/empty after LF or length > 200000 chars; if drops > 5%, stop. Official WT103-raw splits. Frozen P1.1 reconstructed_article_70_15_15: train SHA256 2b0474c5700dc1eba14def572aa23cc227e4c59c10c2de3ce6b7bda75d137687; val SHA256 4d51644b84d05050bfc8c515079e60f6e437082b6cce2122e9ed00e7b1db2b1c; test SHA256 3bd193458f4c494d84dae345548c0c01cb6cd7275e98d6ed39a41d517a093baf. No split substitution.

Exclude NaN/Inf; ClimbMix; val/test in train; new tokenizer after BPB; ratio -1; HF Trainer; writing into p1-fixed-d20-3x or p2-en-then-tl; P1.1/P2 weights as parent; attention patch; test/weight-path leak without a dated deviation card. Do not drop an arm because a contrast looks bad. d4 smoke is not confirmatory.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Tagalog: frozen P1.1 reconstructed_article_70_15_15 train, no rebuild. English: Salesforce/wikitext, wikitext-103-raw-v1, official splits, re-download.

TL0: one seed, d8 and d20, $N_{TL0} = \text{ceil}(3 \cdot T_{tl_train} / 65536)$, $B=65536$, $T=2048$.

B1/B2/B3: one seed each at d20, $N=294$, $D_{\text{phase2}}=19267584$.

After val seal: test_access_count=0 until one B2-only touch (two component reads) on the Q3 legacy holdouts. B1 and B3 are not tested.

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

Q1 is "It's complicated": public corpora and P1.1/P2 outcomes exist; no P3 tokenizer, TL0, B0-B3, val_bpb_full, or test_bpb has been computed.

P3 is a separate preregistration and does not amend #306780 or #306935. It has overlapping source observations with them through named frozen P1.1 Tagalog artifacts and named legacy holdouts, but it has fresh P3 weights, a new P3 tokenizer, new P3 training trajectories, and no previously observed P3 outcomes.

Lockbox: P3 scalars release after val seal plus the B2-only test event, except the blocked-P0-T path. One-person lab: encrypted lockbox with dated access audit.

Protocol SHA256 899ba83f0b36f2b4bf4c16b3c675e58788d7763cb439f8a8c3a3c061bda2b986. Pin 92d63d4. Test files stay outside train dirs.

BUNDLE

This pre-registration is part of a bundle which includes:

#306,780 - <https://aspredicted.org/6r6v4v.pdf> - Title: 'NANOCHAT-FILIPINO P1.1 - WikiText-TL-39 fixed-budget depth vs held-out BPB'